

M53: LDA Switch-Count Diagnostics

Why does count_topic diverge from count_time and count_typing?

Final M52 sample: N = 101 participants

Pairwise correlations on raw counts (z-equivalent):

Time x Typing : $r = +0.596$ (the two that "go together")

Time x Topic : $r = +0.556$

Typing x Topic: $r = +0.504$

Residual flag rule: $|z_{\text{topic}} - \text{mean}(z_{\text{time}}, z_{\text{typing}})| \geq 1.5 \text{ SD}$

Participants flagged: 11 / 101

Top suspects deep-dived in pages 4-5: 8 participants (highest |residual|).

Reading guide:

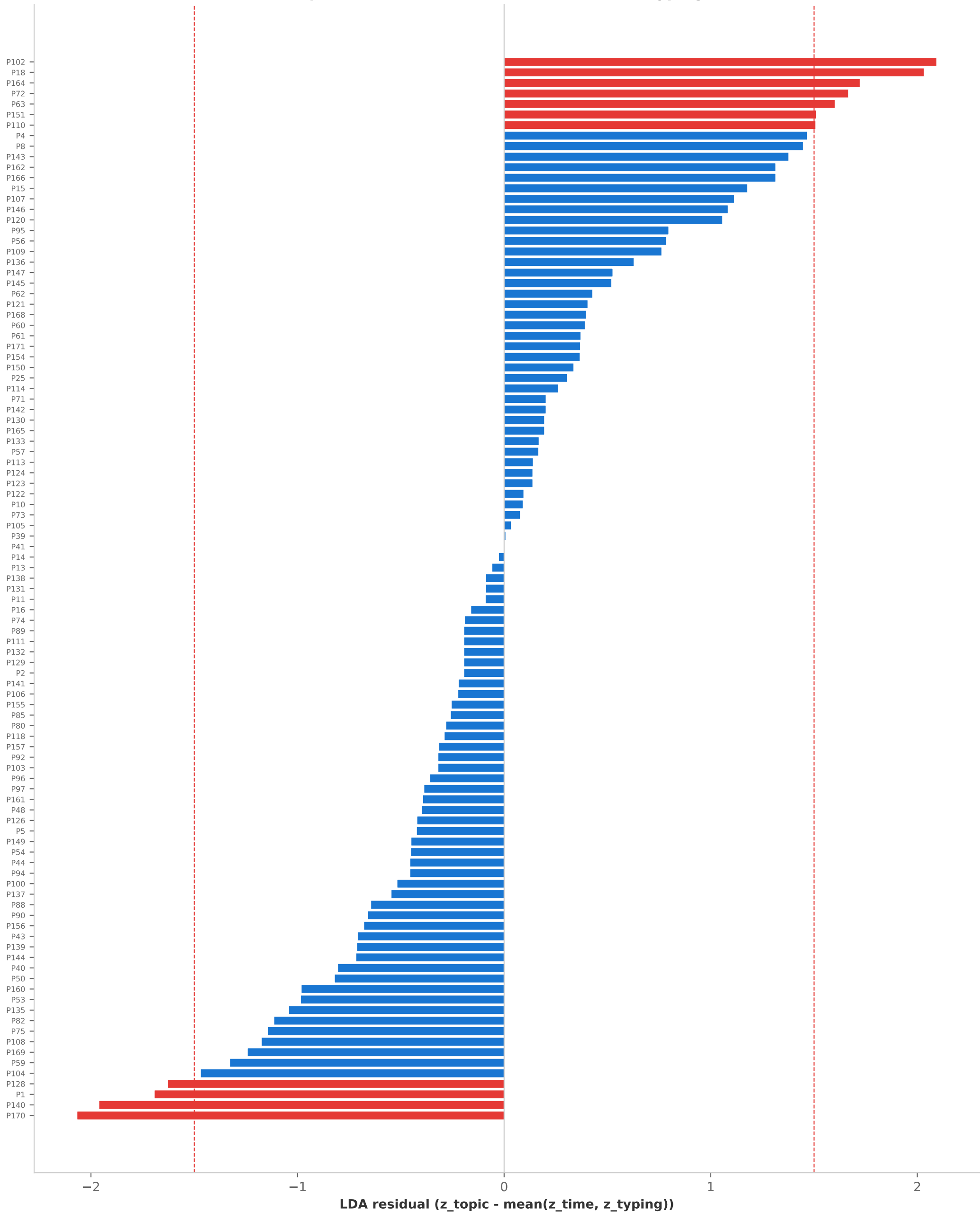
Page 2 - LDA residual ranking (which participants drive the divergence)

Page 3 - Pairwise scatter plots of z-scored counts

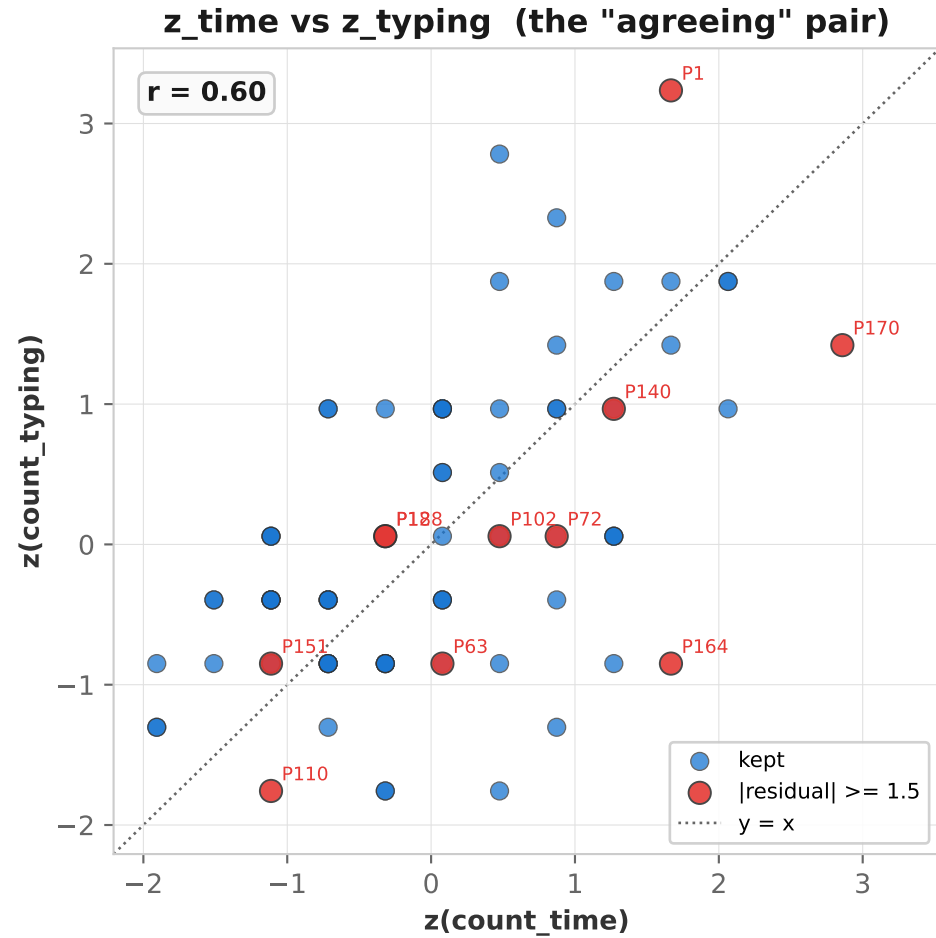
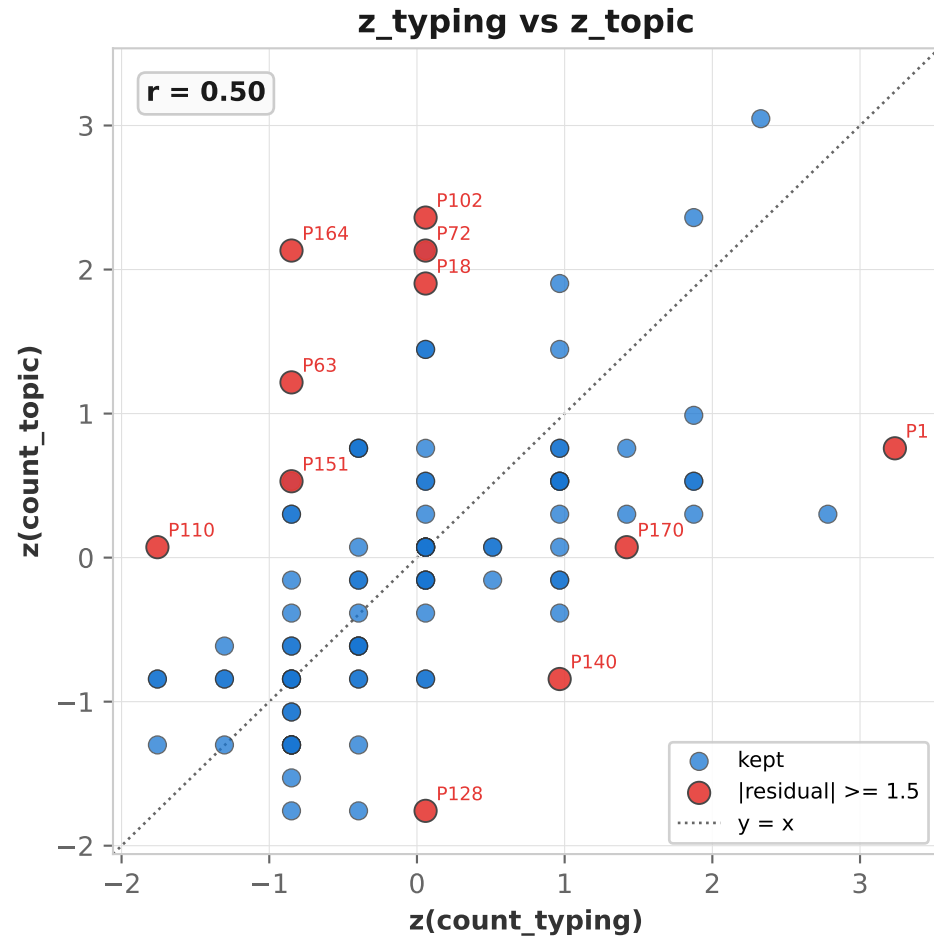
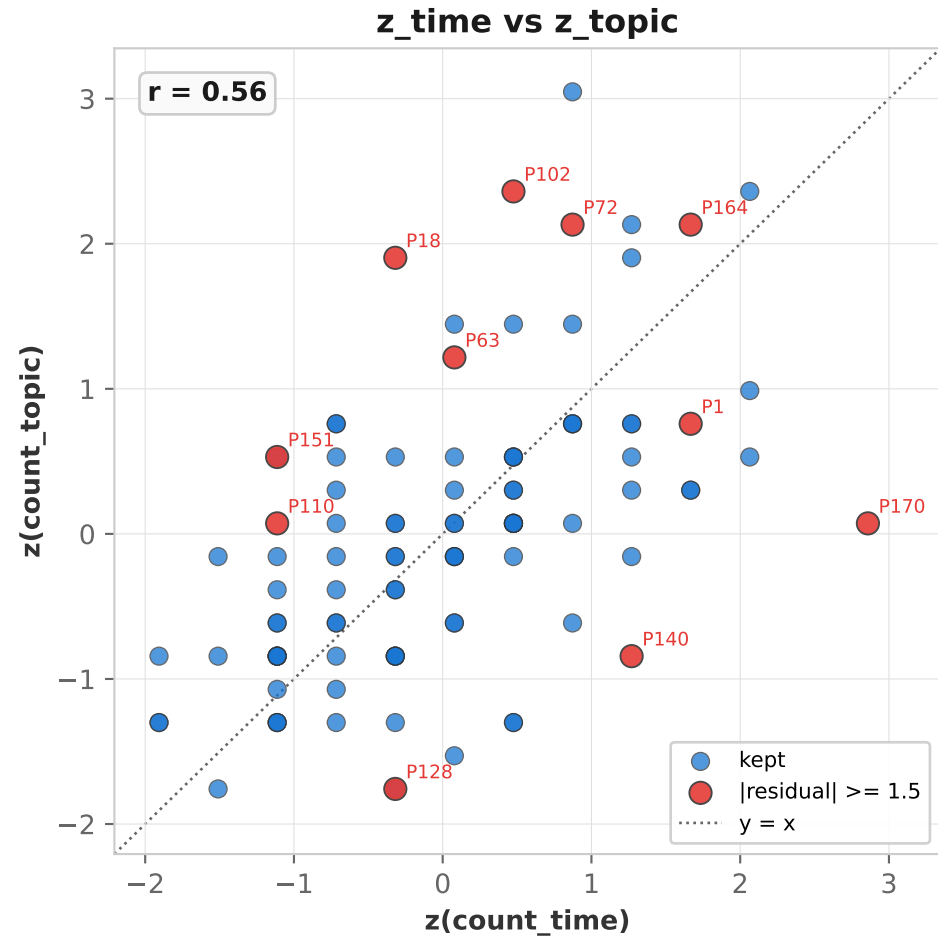
Page 4 - count_topic vs avg n_pages (computational stability)

Page 5 - Page-by-page topic sequences for the top suspects

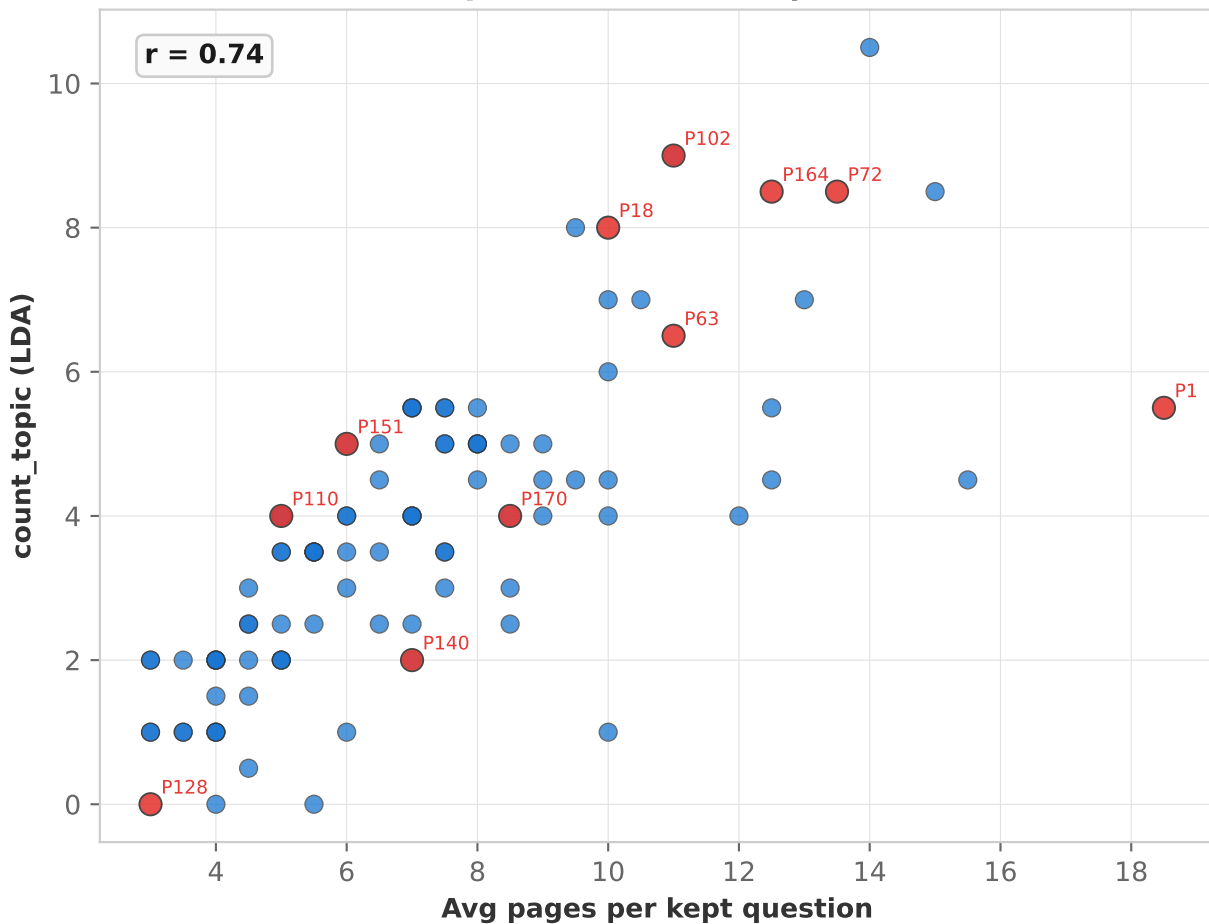
LDA Residual by Participant
(positive = LDA inflated relative to time/typing)



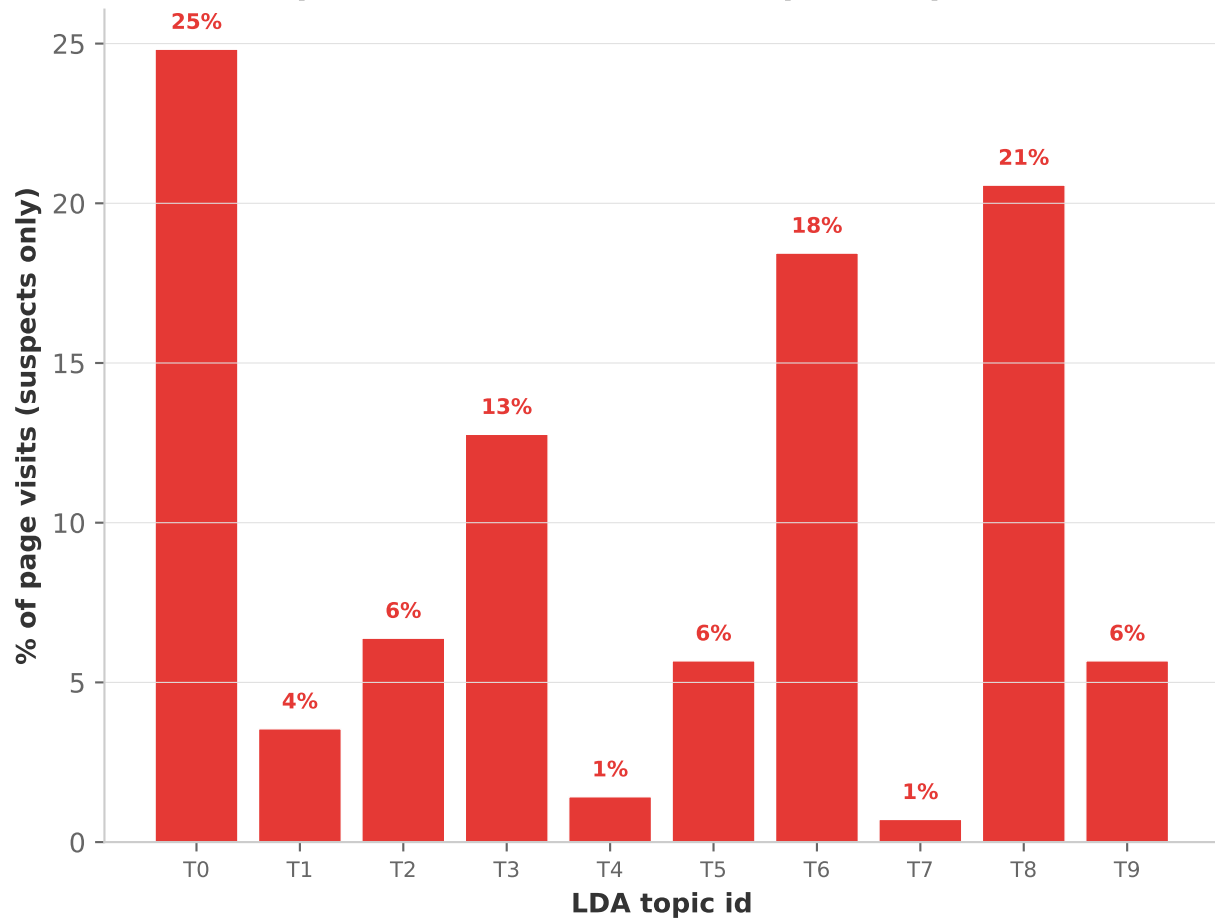
Pairwise Z-Score Scatter Plots



**LDA Switches vs Page Count
(computational stability check)**



Topic Distribution Across Suspect Sequences



TOP LDA-RESIDUAL PARTICIPANTS - PAGE SEQUENCES

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Notation: topic = LDA topic id (0..9) | time: E=exploit (>60s) X=explore (<=60s)
typing: T=typing/paste on page -=-none

P102 | trial 1 (art_history) | pages=11 switches time=3 topic=9 typing=2
topics: 0,8,0,-1,0,8,0,0,8,9,8
time: XXEXXXXXXEE
typing: -TT-----

P102 | trial 2 (psychology) | pages=4 switches time=0 topic=3 typing=3
topics: 1,6,3,-1
time: EEEE
typing: -T-T

P170 | trial 1 (psychology) | pages=8 switches time=4 topic=5 typing=4
topics: 3,3,3,6,1,5,-1,5
time: XEXEEEX
typing: -T-T----

P170 | trial 2 (art_history) | pages=9 switches time=8 topic=3 typing=3
topics: 8,7,0,5,5,5,5,5,5
time: EXEXEXEXE
typing: T-T-----

P18 | trial 1 (art_history) | pages=1 switches time=0 topic=0 typing=0
topics: 9
time: E
typing: T

P18 | trial 2 (psychology) | pages=10 switches time=2 topic=8 typing=2
topics: 6,3,6,6,1,3,6,3,6,3
time: EXXXXXXEE
typing: T-----TT

P140 | trial 1 (psychology) | pages=7 switches time=4 topic=2 typing=3
topics: 6,6,6,3,6,6,6
time: EXXXEXE
typing: ----T-T

P140 | trial 2 (art_history) | pages=2 switches time=1 topic=1 typing=1
topics: 9,8
time: XE
typing: -T

P164 | trial 1 (psychology) | pages=7 switches time=3 topic=2 typing=2
topics: 6,3,6,6,6,6,6
time: XEEXXXE
typing: ----T--

P164 | trial 2 (art_history) | pages=18 switches time=6 topic=15 typing=0
topics: 0,9,0,8,8,4,8,8,0,8,0,9,8,9,8,-1,8,0
time: XXXXXXXXXXXXEXXXXXX
typing: -----

P1 | trial 1 (art_history) | pages=25 switches time=3 topic=4 typing=5
topics: 8,8,0,0,0,8,8,8,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,9
time: XXXXXXEXXXXXXXXXXXXXXE
typing: -----T-----T-----T

P1 | trial 2 (psychology) | pages=12 switches time=6 topic=7 typing=6
topics: 1,2,6,2,6,6,6,2,1,2,2,2
time: XXXEXXXEXEXX
typing: --TT---T-T--

P72 | trial 1 (art_history) | pages=16 switches time=4 topic=11 typing=3
topics: 8,8,0,8,0,8,0,8,8,9,8,8,-1,8,8,0
time: XXXXXXXXXXXXEXEX
typing: -----T--T

P72 | trial 2 (psychology) | pages=11 switches time=3 topic=6 typing=1
topics: 3,3,2,3,2,3,3,3,2,3
time: XXEXXXXXXEE
typing: -----T

P128 | trial 1 (psychology) | pages=3 switches time=2 topic=0 typing=2
topics: 6,6,6
time: XEX
typing: -T-

P128 | trial 2 (art_history) | pages=2 switches time=0 topic=1 typing=0
topics: 4,0
time: EE
typing: TT
